

A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech with Intelligent Symptom Capture, Risk Assessment, Clinical Documentation, and EHR Integration

ABSTRACT

Patients in Bangladesh often struggle to describe symptoms to the doctors clearly specially in regional dialects or Banglish, leaving incomplete or rushed consultations. This project proposes a voice-only pre-screening system where the patient speaks naturally in Bangla, Banglish, or in a regional dialect; the system normalizes and transcribes the speech, extracts clinical entities, asks targeted spoken follow-up questions to close information gaps, and compiles an explainable structured pre-screening report before the doctor enters. A rule-based red-flag check inside the risk-assessment module discloses life-threatening symptoms prominently for the doctor. The system shortens consultation time, reduces documentation burden, and improves information completeness. The pipeline processes speech through sequential stages of transcription, normalisation, and clinical entity extraction, after which a missing-information module flags incomplete details and drives the conversational follow-up loop until the patient's case is sufficiently documented. This closes gaps that inconsistent audio quality and code-switched speech would otherwise leave unresolved. The explainable risk-assessment layer produces a plain-language justification alongside each severity level, allowing the physician to verify the system's reasoning rather than accept an opaque score. The final report is written directly into the patient's electronic health record, linking the pre-screening summary, transcript, and risk profile to the same encounter for future reference. By handling routine symptom elicitation before the consultation begins, the system is intended to give physicians in high-volume clinics more time for examination and clinical decision-making, while extending consistent, dialect-aware pre-screening to patients who are often underserved by English-first digital health tools. The design targets low-resource deployment, using lightweight, open components suited to clinics with limited computing infrastructure, so the approach remains feasible for wider rollout across Bangladesh.